WORKED SOLUTION

STEP II 2020, Question 11

Probability and Statistics • Runs in Coin Tossing

STEP II 2020, Question 11

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

The game has very little memory: all that matters at any moment is which face came up last and how long the current run is. In particular, as soon as the face changes, the game is in exactly the same position as if that new face had been the very first toss. So we condition on the first toss and write the probability that A wins, given a head first, in terms of the probability that A wins, given a tail first, and vice versa. That gives two simultaneous linear equations.

Solution

Throughout, $0 < p < 1$. (If $p = 0$ or $p = 1$ every toss is the same, the game ends on the second or third toss and the answers are trivial.) Write

$$\alpha = \mathrm{P}(\text{A wins} \mid \text{first toss H}), \qquad \beta = \mathrm{P}(\text{A wins} \mid \text{first toss T}).$$

Tosses are independent, so what happens after any toss depends only on the tosses still to come.

Part (i)

The game ends. The game is still going after n tosses only if no two consecutive tosses so far are the same, that is, the first n tosses alternate: $HTHT\ldots$ or $THTH\ldots$. For $n = 2m$ each of these two sequences contains m heads and m tails, so

$$\mathrm{P}(\text{not ended after } 2m \text{ tosses}) = 2p^m q^m = 2(pq)^m \leq 2\left(\tfrac{1}{4}\right)^m,$$

since $pq = p(1-p) = \frac{1}{4} - \left(p - \frac{1}{2}\right)^2 \leq \frac{1}{4}$. The event "the game never ends" is contained in every one of these events, so its probability is at most $2\left(\frac{1}{4}\right)^m$ for every m, and letting $m \to \infty$ shows that it is 0.

Given a head first. Look at the second toss.

- With probability p it is a head, and A wins at once.
- With probability q it is a tail. The head before it is now irrelevant: B needs one more tail, A needs two heads in a row, exactly as if this tail were the first toss. So A then wins with probability β.

Similarly, if the first toss is a tail, B wins at once if the second is a tail, and if the second is a head we are in the "head first" position. Hence

$$\alpha = p + q\beta, \qquad \beta = p\alpha.$$

Substituting the second equation into the first, $\alpha = p + pq\,\alpha$, so

$$\alpha = \frac{p}{1 - pq},$$

as required ($1 - pq \geq \frac{3}{4} > 0$). Then $\beta = \dfrac{p^2}{1 - pq}$.

Overall. Conditioning on the first toss,

$$\mathrm{P}(\text{A wins}) = p\alpha + q\beta = \frac{p^2 + p^2 q}{1 - pq} = \frac{p^2(1+q)}{1 - pq}.$$

Part (ii)

Now α and β refer to the game needing runs of three. Suppose the first toss is a head.

- HH then H, probability p^2: A wins.
- HHT, probability pq, or HT, probability q: the first tail since the start has just appeared and the earlier heads no longer count, so we are in the "tail first" position and A wins with probability β.

These cases cover everything, so

$$\alpha = p^2 + (q + pq)\,\beta,$$

which is the given result. Swapping the roles of the two faces (in the "tail first" position, TTT gives B the win and contributes 0),

$$\beta = (p + pq)\,\alpha.$$

Ask yourself How do the factors $q + pq$ and $p + pq$ turn into $1 - p^2$ and $1 - q^2$?

Since $p + q = 1$, we have $q + pq = q(1 + p) = (1 - p)(1 + p) = 1 - p^2$, and similarly $p + pq = p(1 + q) = 1 - q^2$. This also makes sense directly: $1 - p^2$ is the chance that the two tosses after an opening head are not both heads, which is exactly when the run is broken. So the equations are

$$\alpha = p^2 + (1 - p^2)\beta, \qquad \beta = (1 - q^2)\alpha,$$

and substituting gives $\alpha\left[1-(1-p^2)(1-q^2)\right]=p^2$, that is,

$$\alpha=\frac{p^2}{1-(1-p^2)(1-q^2)}.$$

(The denominator is positive, because $0<1-p^2<1$ and $0<1-q^2<1$.) Finally,

$$\mathrm{P}(\text{A wins})=p\alpha+q\beta=\alpha\big(p+q(1-q^2)\big)=\alpha\left(p+q-q^3\right)=\alpha\left(1-q^3\right),$$

so

$$\mathrm{P}(\text{A wins})=\frac{p^2(1-q^3)}{1-(1-p^2)(1-q^2)},$$

as required.

Part (iii)

The same argument works for runs of length a and b. Given a head first, either the next $a-1$ tosses are all heads (probability p^{a-1}) and A wins, or a tail appears among them (probability $1-p^{a-1}$). In the second case the game simply restarts from that tail, which puts us in the "tail first" position. Likewise, given a tail first, B wins with probability q^{b-1}, and otherwise a head turns up before B's run is complete. So

$$\alpha=p^{a-1}+\left(1-p^{a-1}\right)\beta,\qquad \beta=\left(1-q^{b-1}\right)\alpha,$$

and therefore

$$\alpha=\frac{p^{a-1}}{1-\left(1-p^{a-1}\right)\left(1-q^{b-1}\right)}.$$

Again the denominator is positive, since both brackets lie strictly between 0 and 1. Then

$$p\alpha+q\beta=\alpha\left(p+q-q^b\right)=\alpha\left(1-q^b\right),$$

so

$$\boxed{\mathrm{P}(\text{A wins})=\frac{p^{a-1}\left(1-q^b\right)}{1-\left(1-p^{a-1}\right)\left(1-q^{b-1}\right)}}$$

If you prefer, the denominator expands to $p^{a-1}+q^{b-1}-p^{a-1}q^{b-1}$.

The case $a=b=2$. The formula becomes

$$\frac{p(1-q^2)}{1-(1-p)(1-q)}=\frac{p(1-q)(1+q)}{1-pq}=\frac{p^2(1+q)}{1-pq},$$

using $1-p=q$, $1-q=p$ in the denominator and $1-q=p$ in the numerator. This agrees with part (i). Putting $a=b=3$ gives back part (ii) exactly.

Checks

- *Fair coin, part (i).* With $p=q=\frac{1}{2}$ the answer is $\frac{1}{4}\cdot\frac{3}{2}\big/\frac{3}{4}=\frac{1}{2}$, as symmetry demands. The same happens in (iii) whenever $a=b$ and $p=\frac{1}{2}$.

- *Extremes.* As $p \to 1$ the answer in (iii) tends to 1, and as $p \to 0$ it tends to 0, as it should.
- *Numbers.* With $p = 0.4$ the part (i) answer is $0.16 \times 1.6/0.76 \approx 0.3368$, and the part (ii) answer is ≈ 0.2713. A computer simulation of 200 000 games gave 0.336 and 0.271. For $p = 0.55$, $a = 4$, $b = 2$ the formula gives 0.2450, again matched by simulation.

Common slips

- Forgetting that once the face changes, the earlier run counts for nothing. That's exactly why the position after a change is the same as "that face first".
- In (ii), missing the case HHT: the run can be broken on the second *or* the third toss.
- Giving α as the final answer to "the probability that A wins". You still have to average over the first toss, $p\alpha + q\beta$.

Beyond STEP

This is a first-step analysis of an absorbing Markov chain. The states are "current face and length of run", and the absorption probabilities satisfy linear equations obtained by conditioning on the next step. The same technique gives expected waiting times for patterns in coin tossing, and leads on to Penney's game, where the order of the patterns produces some surprising non-transitive results. Grimmett and Welsh, *Probability: An Introduction*, covers Markov chains and absorption at the right level.

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